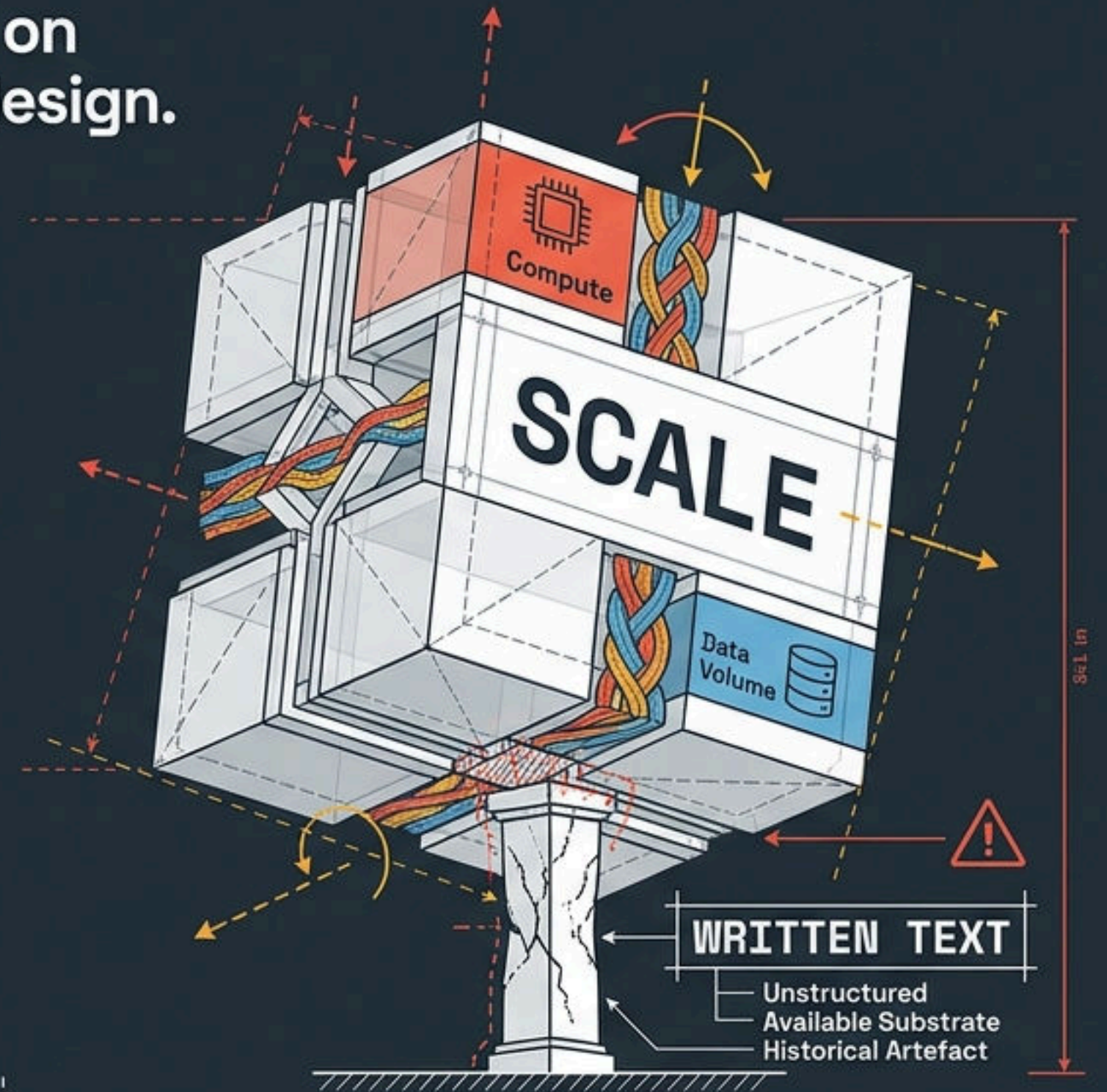
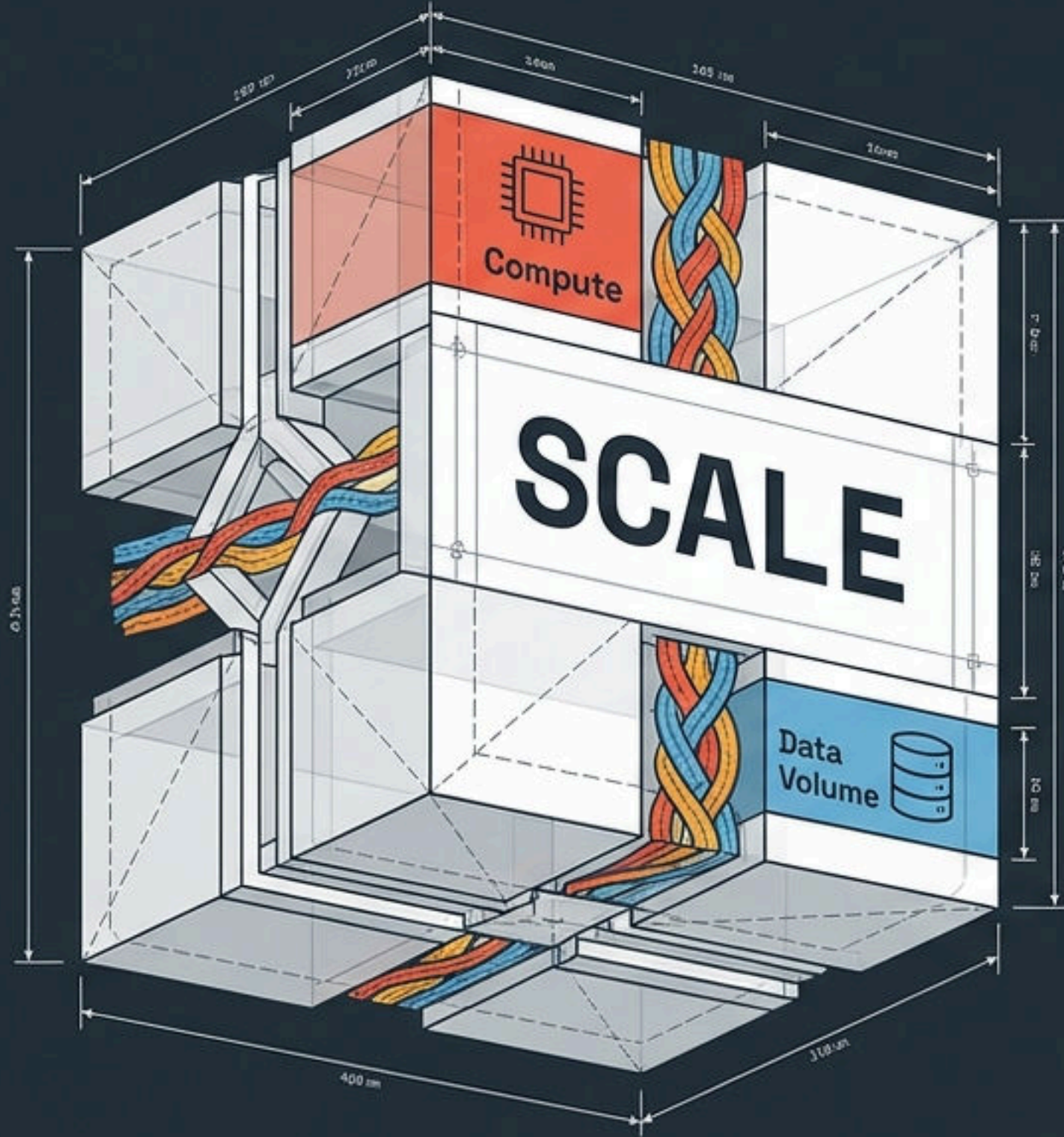


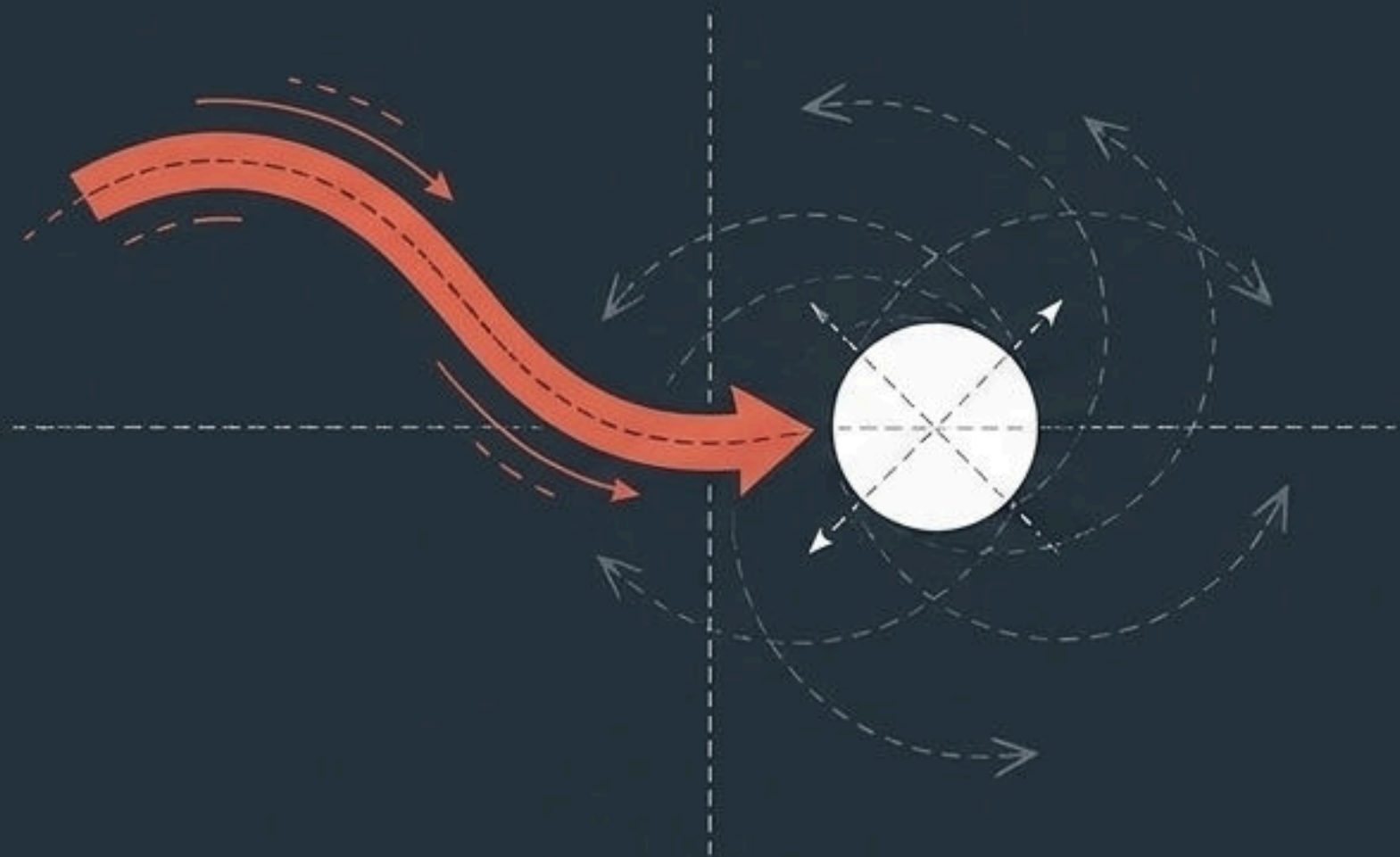
The foundational choice of modern AI is based on historical accident, not principled curriculum design.



- Large Language Model development aggressively optimizes **compute**, tokenization, attention variants, and alignment.
- Underneath sits an **unexamined assumption**: that models should learn primarily from **written text**.
- Text is abundant, cheap, and digitized at scale. But this makes it an **AVAILABLE** substrate, not necessarily an **OPTIMAL** one.

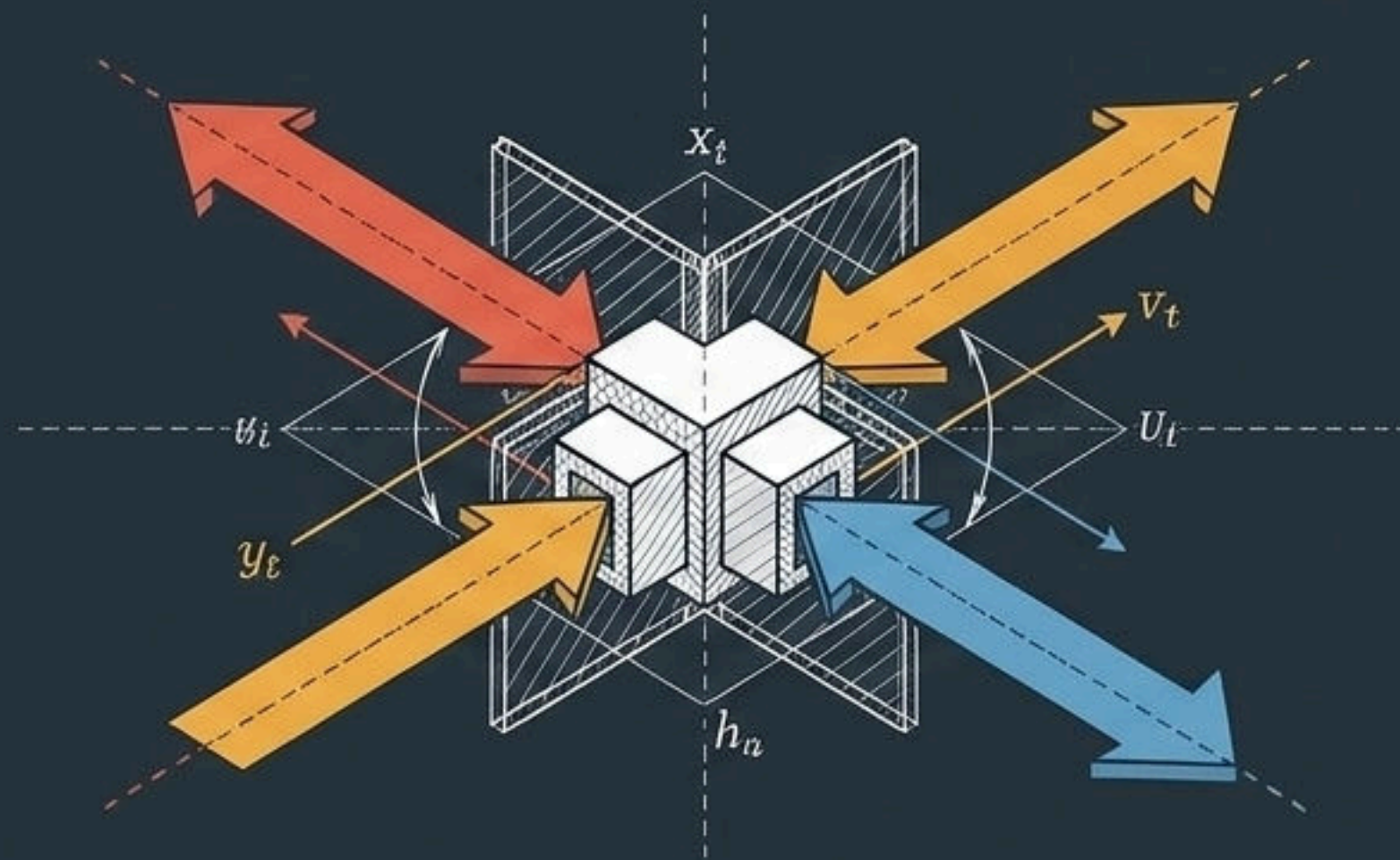
The Representational Constraint Hypothesis

A Cue (Single Channel)



A single channel supplies a loose cue. It may or may not be exploited, leaving the representation untethered.

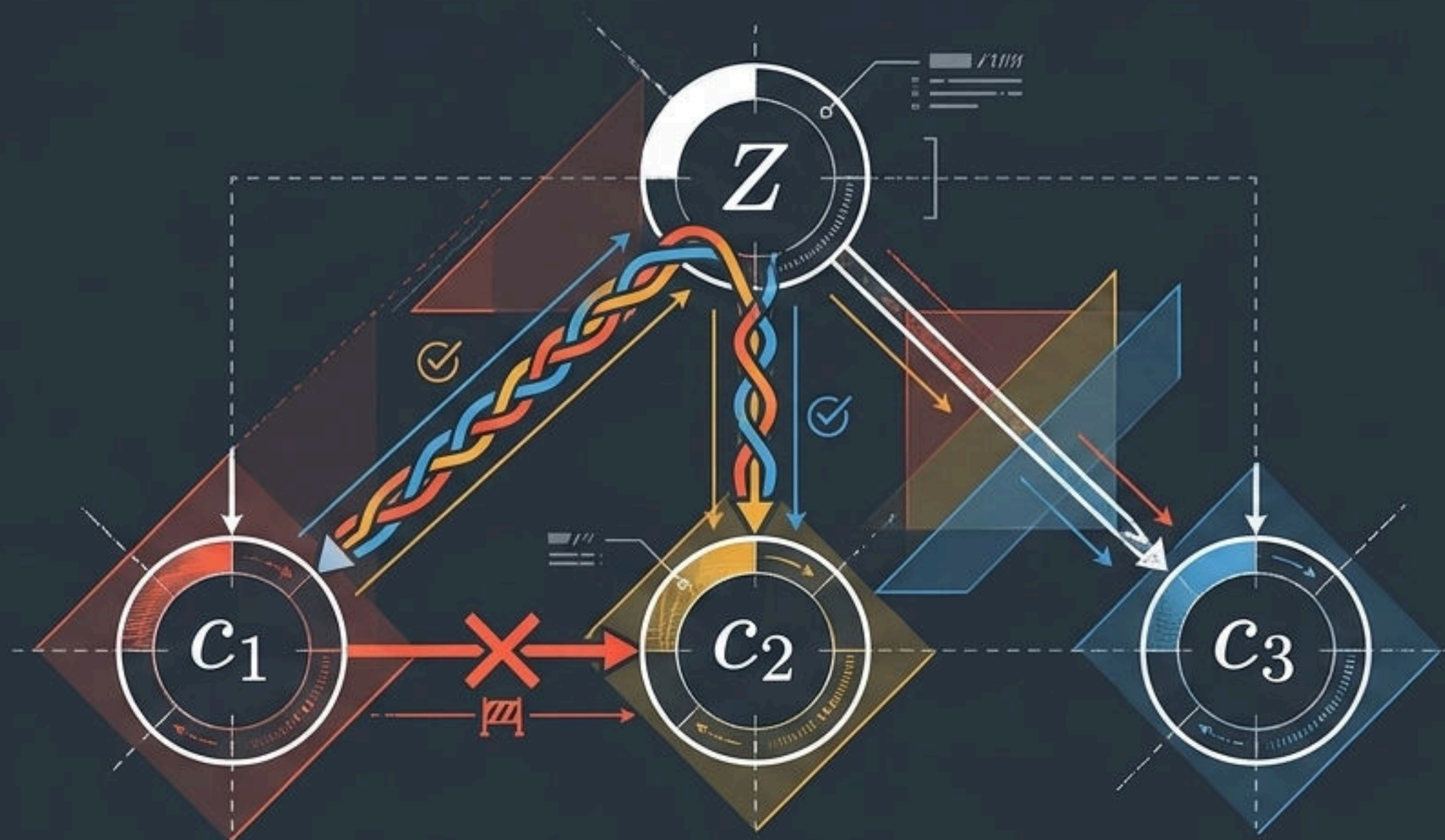
A Constraint (Multi-Channel)



Multiple independent channels supply a structural constraint. They must be jointly satisfied, forcing the discovery of underlying invariants.

Foundational curricula matter most when they force a learner to reconcile multiple, partially independent views of the same latent state. Agreement between independently corrupted channels acts as the ultimate supervisory signal.

The geometry of cross-channel reconciliation requires passing through the **latent state**.



Let Z denote the hidden variable (the actual event/concept).

Let $C = \{c_1, \dots, c_n\}$ be the observation channels (image, caption, physical contact).

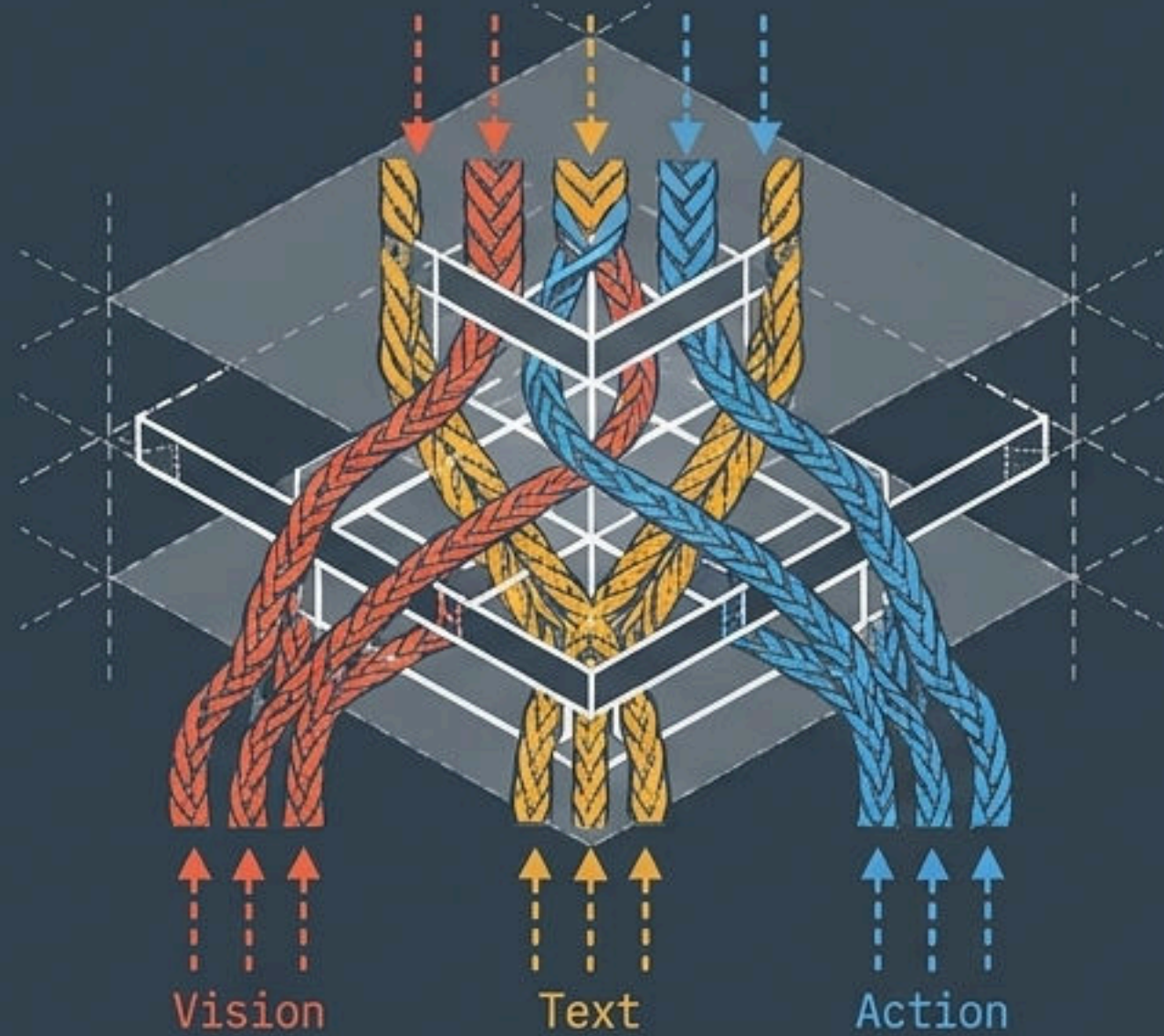
Constraint Diversity (D)

$$D(C) = \sum [I(c_i; c_j) - I(c_i; c_j|Z)]$$

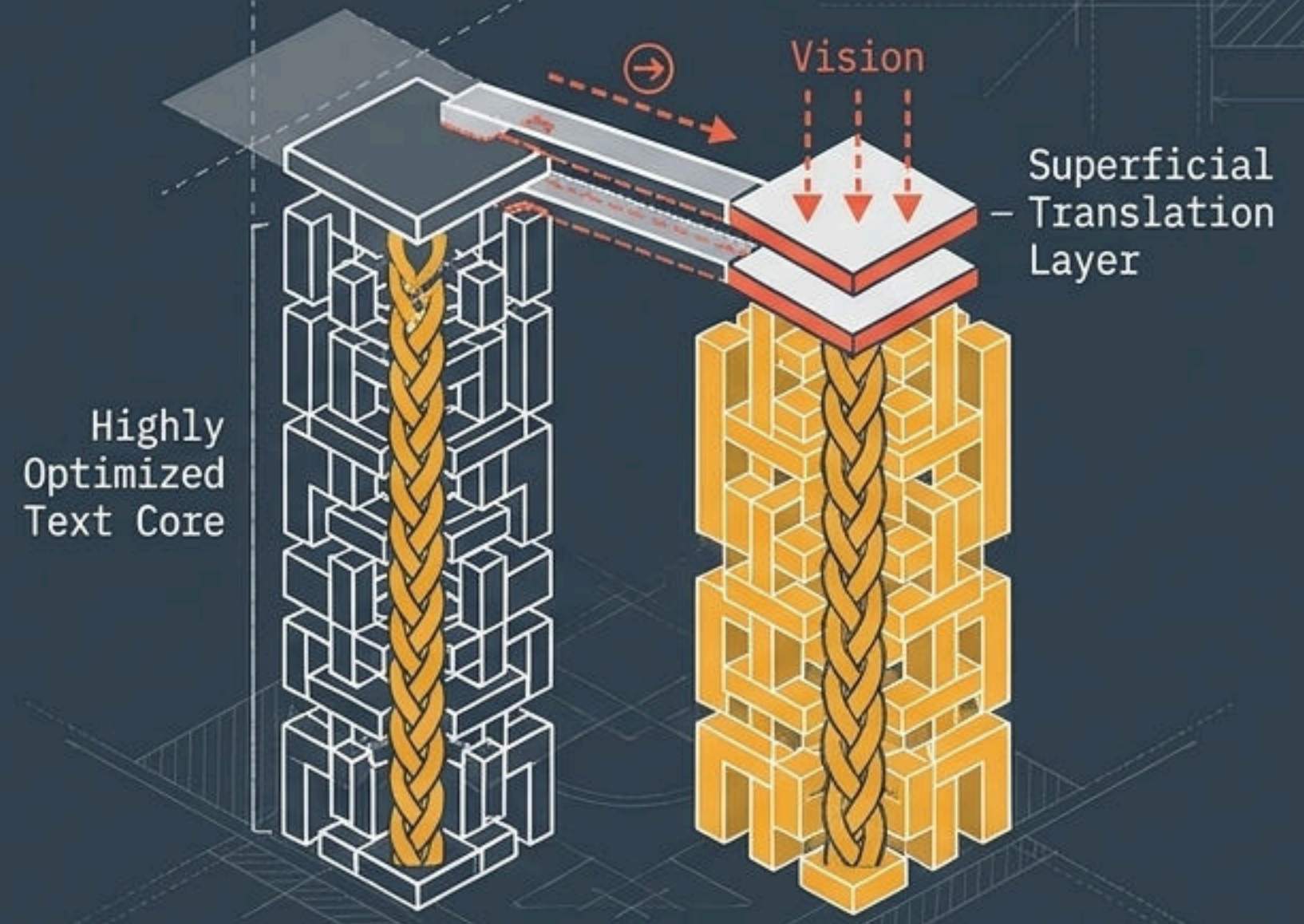
Two channels constrain each other usefully when their agreement is caused by Z rather than surface correlation.

Introducing constraints late triggers the “Translation Shortcut,” bypassing structural reorganization.

Path A: Early Constraint

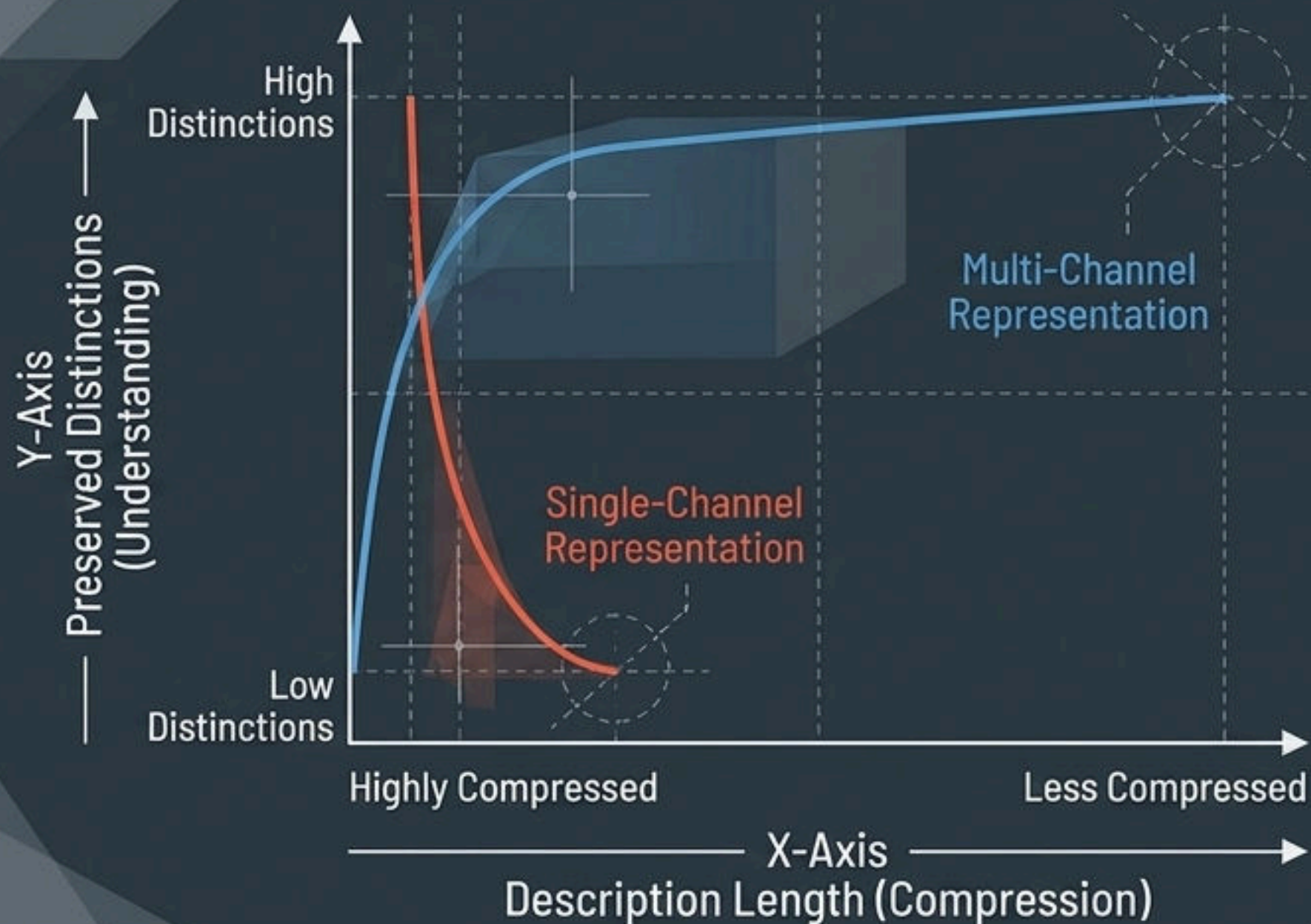


Path B: Late Constraint



- Gradient descent always finds the cheapest path.
- If a model has heavily optimized a single-channel representation, adding a new channel late prompts it to build a cheap mapping (translation) rather than disturb the core.
- To force genuine representational reorganization, constraints must be present when the coordinate system is first built.

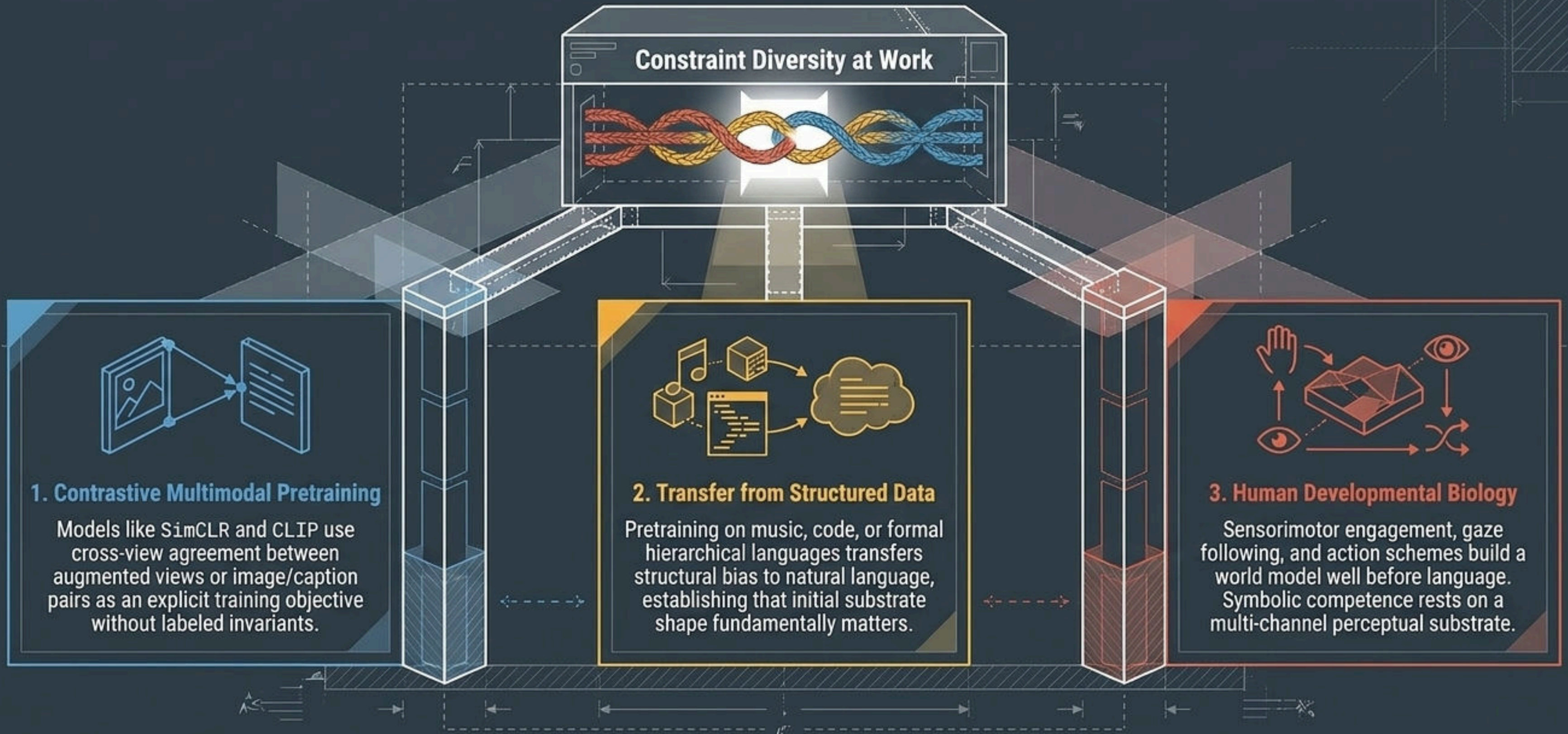
A multi-channel curriculum produces representations that are less compressed, but far richer.



Rethinking the Information Bottleneck:

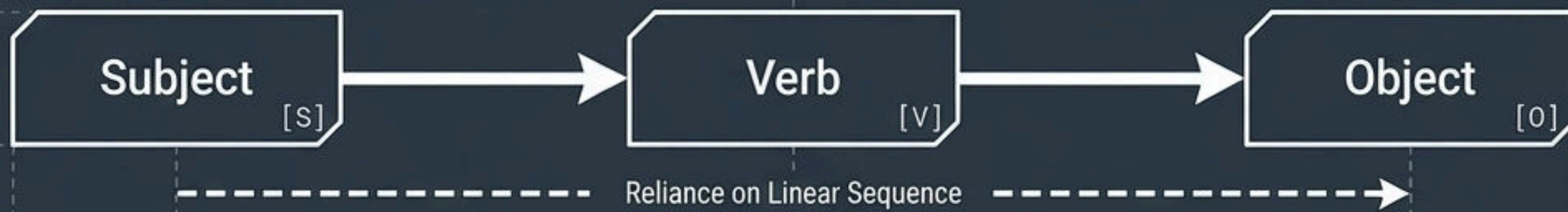
- The prevailing AI assumption is that compression equals understanding. ■
- But a latent organized to render vision, proprioception, and text must retain structure that any single channel would aggressively collapse. ■
- A channel earns its place in a curriculum exactly when it preserves distinctions the other channels discard. ■

Independent discoveries across disciplines converge on Constraint Diversity.

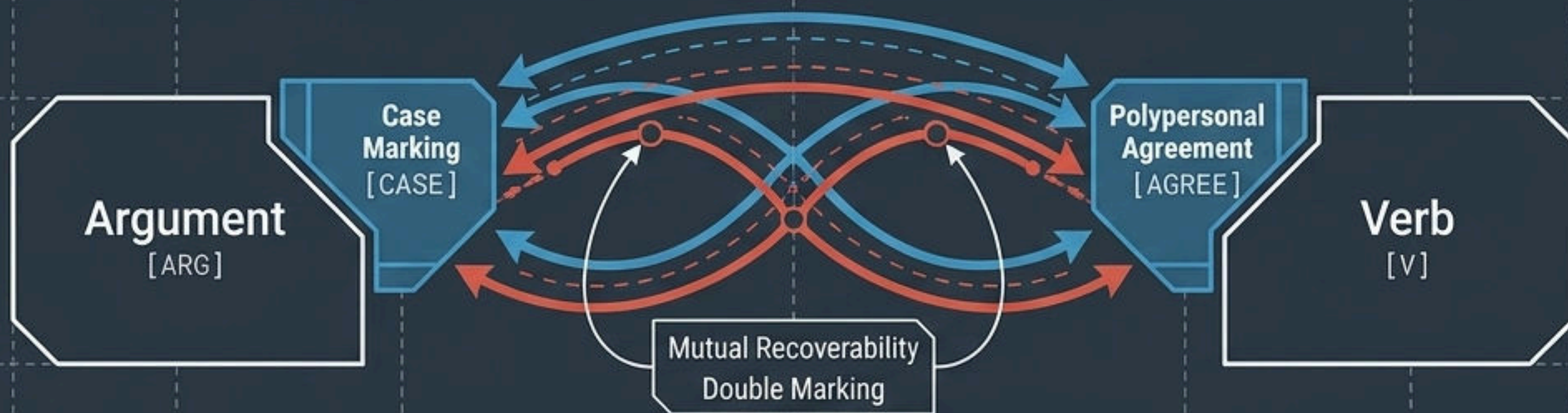


The Basque First Precedent: Redundancy within a channel acts as a constraint.

English: Positional Heuristics

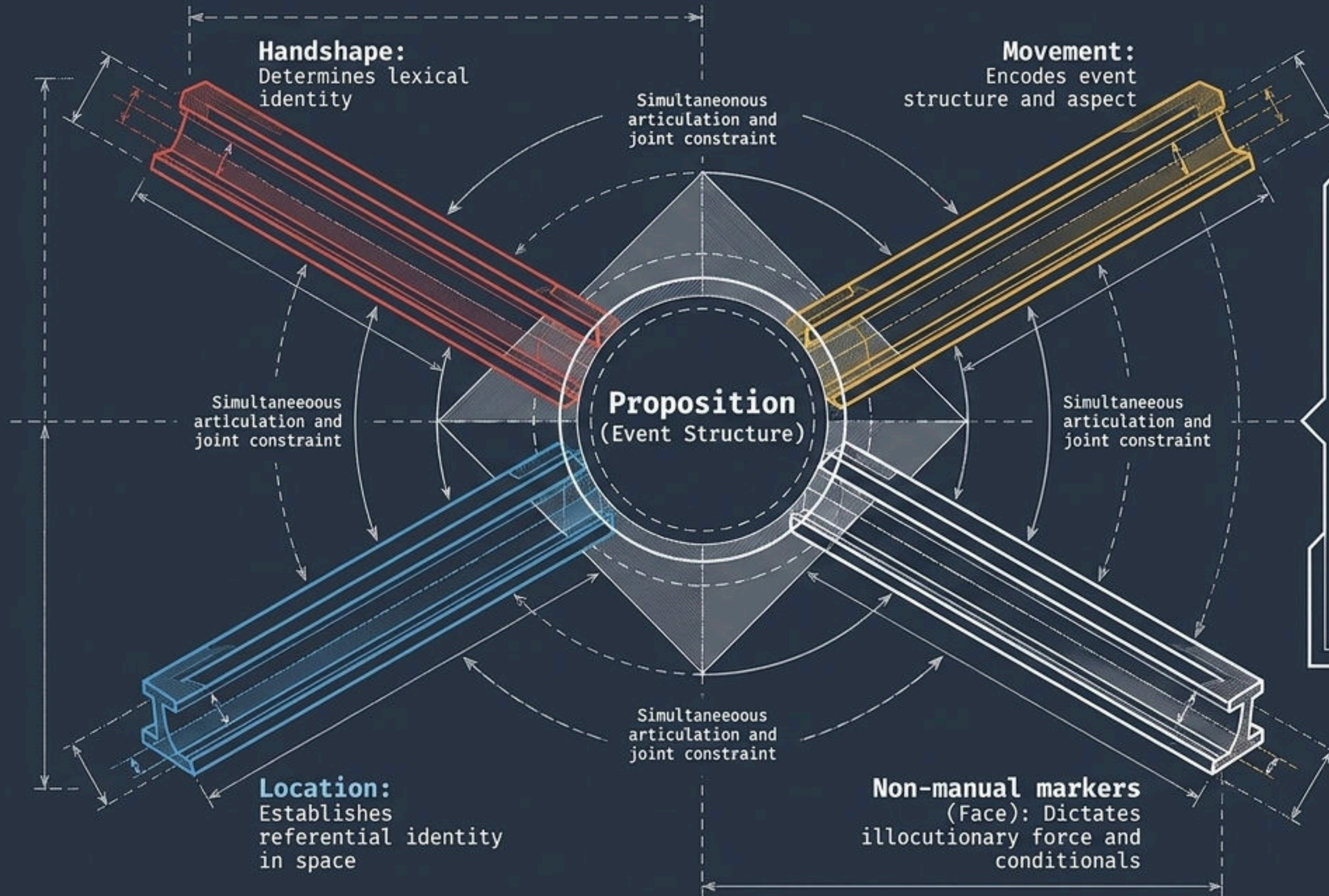


Basque: Double Marking



- Eichler argues that English-first pretraining builds reasoning on a morphologically impoverished foundation.
- Basque marks predicate-argument structure twice (case and agreement).
- This is not about Basque being visually 'richer'—it is about redundancy. The two markings are recoverable from each other, forcing the model to reconcile them and extract the structural relation.

Signed language is the richest naturally occurring high-diversity curriculum.

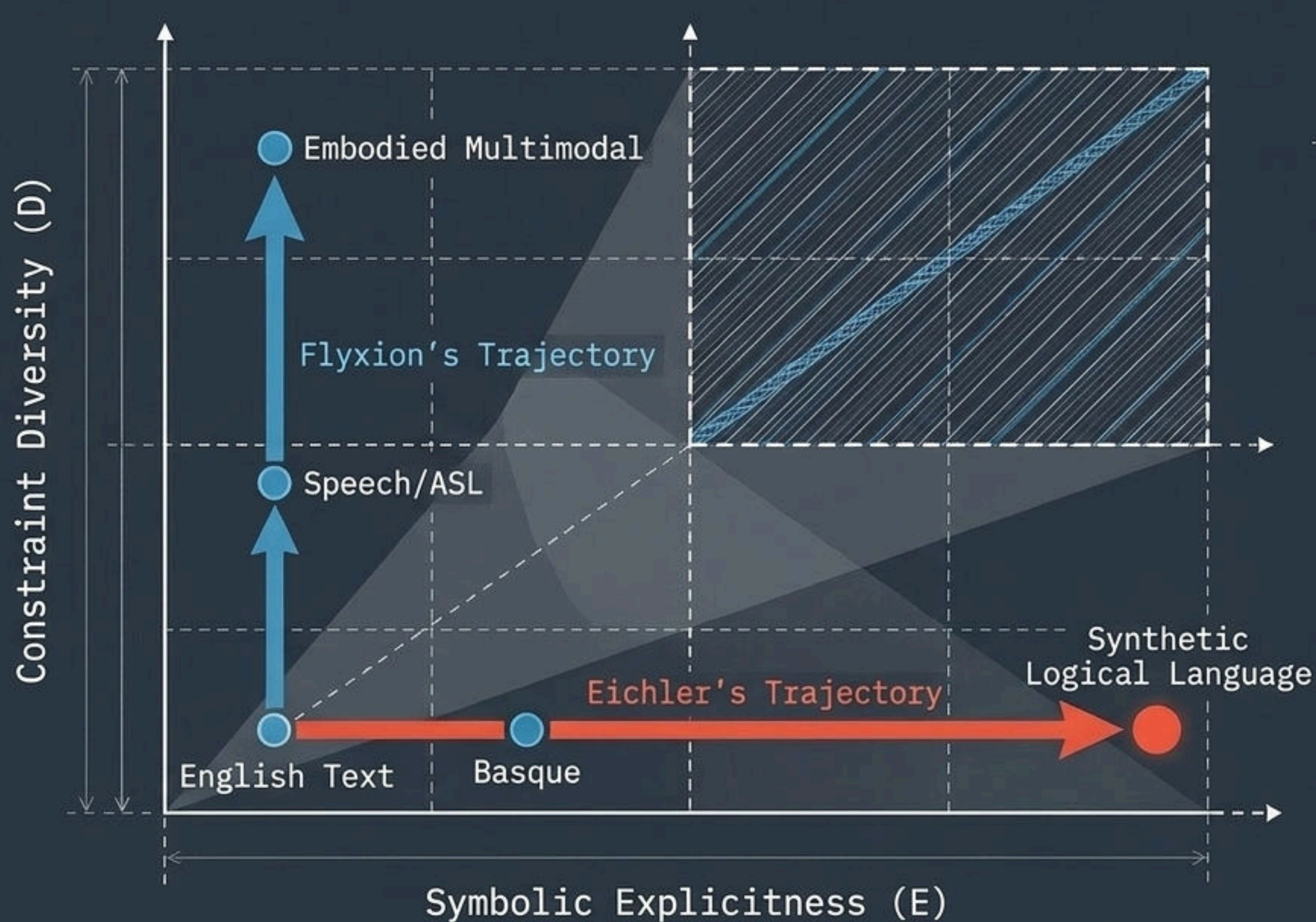


- ASL is a fully grammatical natural language, but its grammar is distributed across simultaneous, mutually constraining **physical channels** rather than serialized into a single string.
- Because these channels are separately observable but jointly constrained by the proposition, predicting one requires **reconstructing** the underlying event.

Comparative Curriculum Diagnostic

Substrate	Modality	Channels	Symbolic Explicitness (E)	Constraint Diversity (D)
English Text	Written	Single	Low	Zero
Basque	Written	Single (Double Marked)	High	Low (but nonzero)
Synthetic Logical	Written	Single	Maximal	Low
American Sign Language	Physical	Multi	Medium	High
Embodied Multimodal	Physical	Multi	Medium	Very High

The Explicitness-Diversity Plane separates competing research trajectories.



- A perfectly explicit synthetic language (max E) minimizes ambiguity but supplies exactly one account of each state—there is nothing to reconcile.
- Multimodal curricula (max D) supply many redundant accounts, but may leave channels individually ambiguous.
- The two programs optimize for different coordinates. Neither subsumes the other.

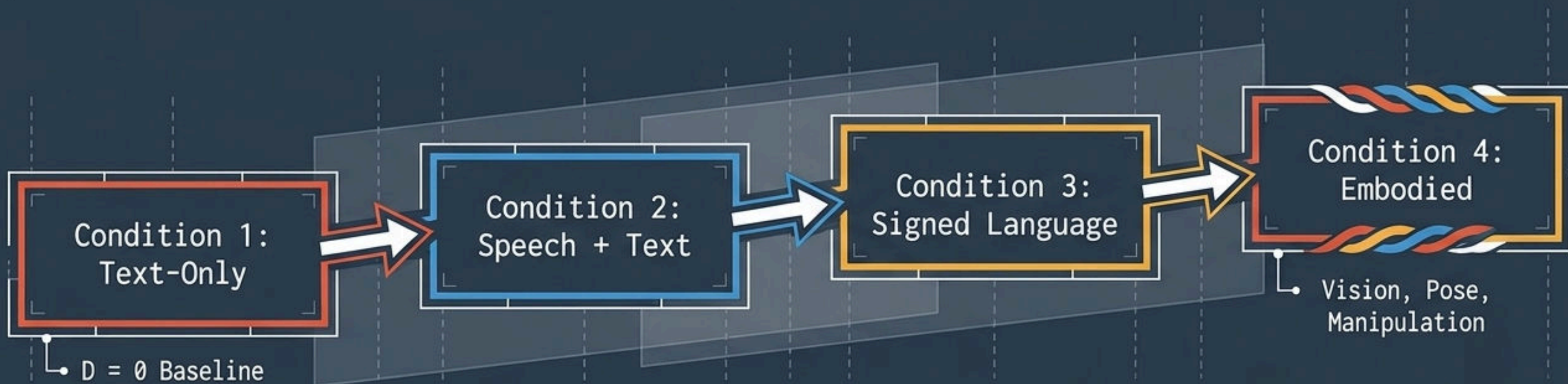
The Unexplored Frontier: The Synthetic Embodied Substrate



What occupies the space of maximal explicitness AND maximal diversity?

- A constructed system where every semantic relation is encoded simultaneously through explicit symbols, physical perception, and action.
- Encodings are mutually recoverable but separately observable, forcing massive constraint satisfaction without pragmatic ambiguity. This is the ultimate, undiscovered foundational curriculum.

The Experimental Roadmap: Isolating the Diversity Variable



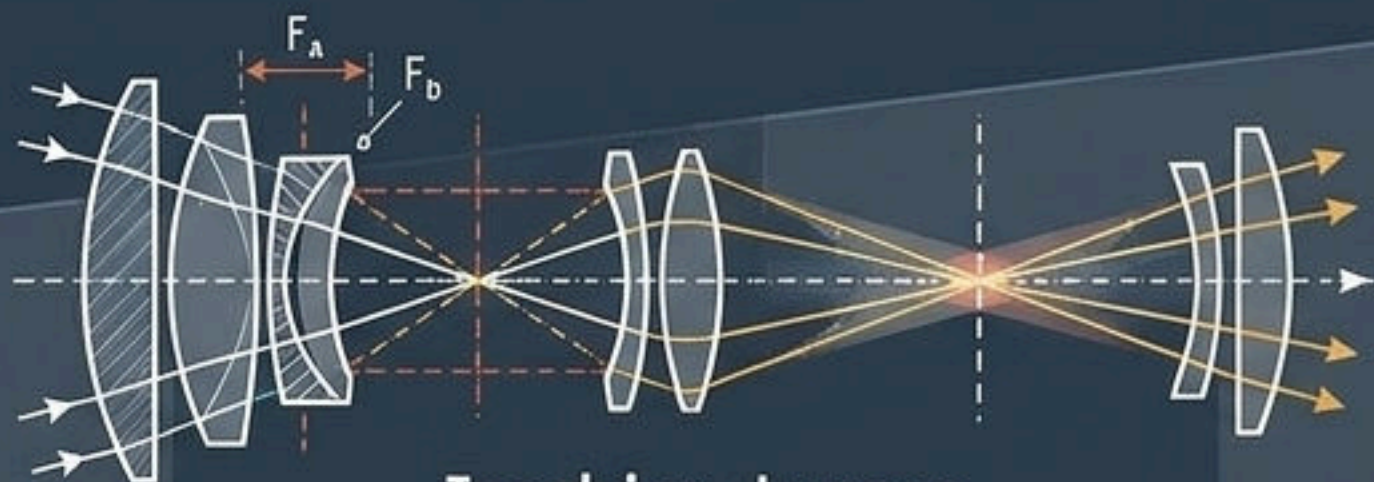
The Setup

- 1-3B parameter scale models.
- Architecture, total data volume, and downstream training held strictly fixed.
- Only Constraint Diversity (D) varies.

The Controls

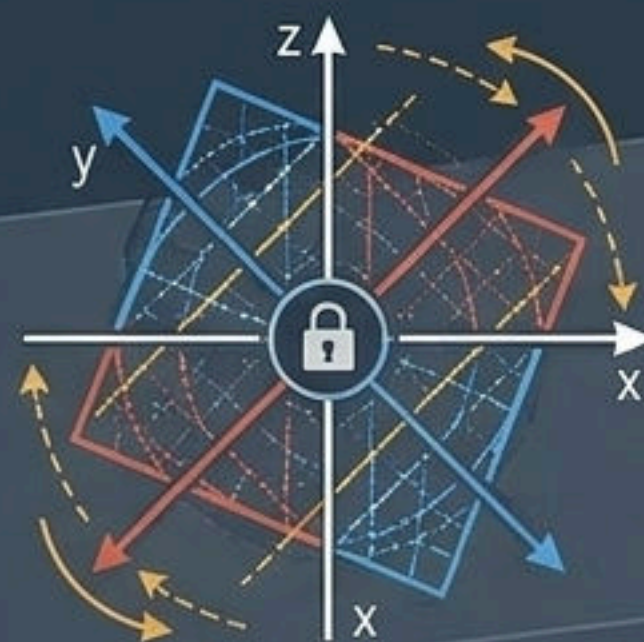
- **Duplicated-Channel Condition:** High correlation, but screens off the latent (proves non-substitutability).
- **Matched-Compute Control:** Ensures diverse constraints aren't just an artifact of compute volume.

Measuring Foundational Geometry, not just benchmark performance.



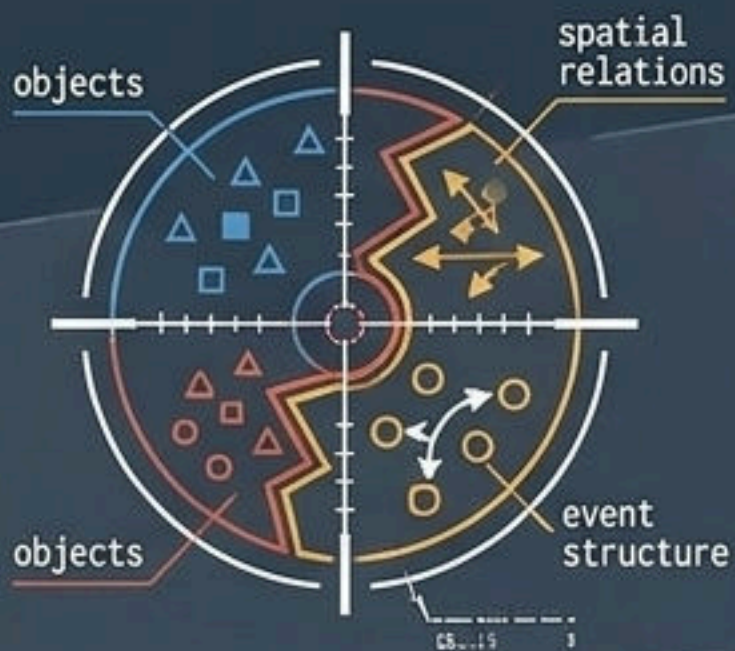
Jacobian Lenses

Identifying the sparse subspace of verbalizable representations.



Centered Kernel Alignment (CKA)

Comparing latent geometry directly across conditions to detect shared organization.



Probing Classifiers

Testing linear recoverability of objects, spatial relations, and event structure.



Held-Out Channel Reconstruction (The Direct Test)

Can the latent accurately reconstruct a withheld sensory channel? If early layers bound the constraints properly, reconstruction succeeds without superficial translation.

The object of foundational curriculum design is not choosing a language or a modality. It is choosing consistency constraints. To build better AI, we must curate the constraints a learner is forced to reconcile in its earliest moments of geometric organization.

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Perception

Text

Action

Symbol